
LEARNING TO ADAPT IN 5-A-SIDE FOOTBALL: FORMATION CHANGES AND ADVANTAGED OPPONENTS UNDER LIMITED VISION

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ABSTRACT

We study cooperative multi-agent learning in a simplified 5-a-side football game with no goalkeepers, where every player sees the opponents and the ball only within a limited radius. Our team is controlled by a single parameter-shared reinforcement learning policy and plays against a fixed, hand-coded heuristic team. We pose three questions of increasing difficulty: (Q1) can the team learn to beat the heuristic opponent, and how does this depend on the formations used; (Q2) can it detect and adapt when the opponent changes formation in the middle of a match; and (Q3) can it detect and adapt to an opponent player who has an advantage, namely full-pitch vision. We formalize the problem as a decentralized POMDP, define measures of adaptation (detection delay, adaptation lag, and regret against an oracle that is told the hidden information), and describe the experiments planned for the final submission.

Keywords Multi-agent reinforcement learning · Football · Partial observability · Formation adaptation · Opponent modelling

1 Introduction

1.1 Motivation

Team sports are among the richest everyday examples of multi-agent decision making. Each player acts on incomplete information (nobody sees the whole pitch at all times), must coordinate with teammates without constant communication, and faces opponents who change their tactics during the match. A coach who notices that the opposing team has switched from a defensive to an attacking shape, or that one opposing player is unusually good at reading the game, will change how the team plays. Building AI teams that can do the same is an open and practical challenge: the same ingredients (partial information, cooperation, adversaries that change) appear in robot soccer [1], warehouse robotics, and multi-drone coordination.

Football has long been used as a testbed for artificial intelligence, from the RoboCup initiative [1] to modern reinforcement learning environments such as Google Research Football [2]. A recent benchmark study shows that when players can observe almost the entire pitch, standard cooperative reinforcement learning methods can beat the built-in bot in 5v5 games with high win rates [3]. That study also names **partial observability** and **robustness to different opponents** as open directions. This project addresses both in a controlled, simplified setting.

1.2 How the Game Works

Before stating the problem, here is the game in plain words. Precise definitions follow in Sections 2 and 3, and all numeric values are still to be decided.

The setting. Two teams of five players play on a rectangular pitch. There are no goalkeepers, so a goal is defended only by the outfield players. The match lasts a fixed number of steps, and the team with more goals at the end wins.

Who controls whom. One team is *ours*: a single learned “brain” (one neural network) is used by all five players, and each player feeds it what it can see and gets back its own action. The other team is the *opponent*: it follows fixed hand-written rules and does not learn.

What a player can do on each step. A player moves, holds its position, passes to a teammate, dribbles toward goal, shoots, or tries to tackle the player with the ball. Passes can be intercepted by defenders near the pass line, tackles can succeed or fail, and shots are more likely to score when close to goal and unmarked.

Formations. A formation is a starting shape, such as 2 defenders, 2 midfielders and 1 forward. Each slot has a home position. Players may leave their home position, but they play worse the farther they are from it, so the formation shapes how the team defends and attacks. Both teams start each match in a formation, and after each goal players return to their home positions for a kickoff. Example formations, written defenders–midfielders–forwards, are 2-2-1, 2-1-2, 1-3-1, 1-2-2 and 3-1-1.

Limited vision. Each player sees the opponents and the ball only if they are within a certain distance of it. It always knows where its own teammates are. This means a player often cannot see the whole opposing team, so it must work out what the opponent is doing from a partial view. A defender can only try to intercept a pass if the ball was within its sight when the pass was played.

What changes during a match.

- *Formation change (Q2)*: the opponent may switch formation mid-match, for example from a defensive shape to an attacking one. Its players walk to their new positions, so there is a short period of disorder. Our team is not told about the change and has to notice it from what it sees.
- *An advantaged opponent (Q3)*: one opponent player may have “full-pitch vision”, meaning it can see every player and every passing lane, and can intercept passes anywhere. Our team is not told which player it is, or whether there is one.

A short walk-through. Suppose the opponent starts in a defensive formation and our team learns to pass patiently and probe for gaps. Halfway through, the opponent switches to an attacking shape. A team that cannot see the whole pitch may keep playing the old way for a while, giving away the ball in dangerous places, until it realizes from the positions it can see that the opponent’s shape has changed. How quickly a team notices and how well it adjusts is what this project measures.

What “learning” means here. The team starts out playing randomly. It plays many matches against the opponent, gets rewarded for goals and winning (and lightly for good passes and tackles), and gradually improves. We then test it in the three situations above.

1.3 Problem Statement

We study a simplified 5-a-side football game with **no goalkeeper**. Two teams of five outfield players play on a 2D pitch. Our team is controlled by a single parameter-shared reinforcement learning policy. The opponent is a fixed, hardcoded heuristic team. Every player sees opponents and the ball only within a limited radius. We ask three questions, in increasing order of difficulty:

1. **Q1 (learning to play)**: Can a team of learning agents beat a fixed heuristic team under limited vision, and how does this depend on the formations used?
2. **Q2 (formation change)**: If the opponent changes formation during the match, can our team detect the change from partial observations and adapt, and how much is lost compared with a team that is simply told the new formation?
3. **Q3 (advantaged opponent)**: If one opponent player has an advantage (it sees the entire pitch), can our team detect and adapt to it?

2 Preliminaries

2.1 Notation

All symbols below are used throughout the report. Numeric values (pitch size, speeds, radius, horizon) are left open and will be fixed during implementation.

Table 1: Notation.

Symbol	Meaning
$\mathcal{N} = \{1, \dots, 5\}$	Players of our team
$\mathcal{M} = \{1, \dots, 5\}$	Players of the opponent team
T	Match horizon (number of steps)
$t \in \{0, \dots, T\}$	Time step
$\mathcal{P} \subset \mathbb{R}^2$	Pitch (a bounded rectangle)
$x_i^t \in \mathcal{P}$	Position of our player i at time t
$y_j^t \in \mathcal{P}$	Position of opponent player j at time t
$b^t \in \mathcal{P}$	Ball position
$\beta^t \in \{\text{ours}, \text{opp}, \text{loose}\}$	Possession state
$\mathcal{F}$	Finite set of formations
$\alpha_f(k) \in \mathcal{P}$	Anchor (home position) of slot k in formation f
f	Our formation (fixed within a match)
$g^t \in \mathcal{F}$	Opponent formation at time t (hidden from our team)
r	Vision radius of a normal player
ρ_j	Vision radius of opponent player j
$\sigma \in \{0, 1, \dots, 5\}$	Index of the opponent player with the full-vision skill ($\sigma = 0$: no skill)
θ	Parameters of the shared policy
G_T	Goal difference (ours minus opponent's) at the end of the match

2.2 Formations and Positional Control

A **formation** $f \in \mathcal{F}$ assigns each of the five players a **slot** k (defender, midfielder, or forward) and an **anchor** $\alpha_f(k)$. Player i occupies slot $k = i$. Formations are not enforced. Instead, a player's effectiveness depends on how far it is from its anchor through the **control rating**

$$c(x, \alpha) = c_{\min} + (1 - c_{\min}) \exp\left(-\frac{\|x - \alpha\|^2}{L^2}\right), \quad (1)$$

where $c_{\min} \in (0, 1)$ and $L > 0$ are constants. Control ratings scale tackle success, pass accuracy, and shot success. When a team switches formation, anchors change immediately and the players must walk to the new anchors, so effectiveness drops temporarily.

2.3 Limited Vision

For a player at position x with vision radius ρ , the **visibility indicator** of a point $z \in \mathcal{P}$ is

$$v(z; x, \rho) = \mathbb{I}[\|z - x\| \leq \rho]. \quad (2)$$

Normal players use $\rho = r$. The **skilled** opponent player uses $\rho = \infty$. Opponents and the ball are observed only when $v = 1$. Teammate positions are always available to our players (a modelling simplification). A defender can attempt to intercept a pass only if the ball was visible to it when the pass was played.

2.4 Decentralized POMDP

A **decentralized partially observable Markov decision process (Dec-POMDP)** [4, 5] is a tuple

$$\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, P, R, \{\Omega_i\}_{i \in \mathcal{N}}, O, \gamma, T \rangle, \quad (3)$$

with state space $\mathcal{S}$, per-agent action sets $\mathcal{A}_i$, transition kernel $P(s' | s, \mathbf{a})$, a shared team reward $R(s, \mathbf{a})$, observation spaces Ω_i , observation kernel $O(\mathbf{o} | s, \mathbf{a})$, and discount $\gamma \in (0, 1]$. Because the opponent is a fixed heuristic, we treat it as part of the environment, so our problem is a Dec-POMDP rather than a two-sided game.

2.5 Policies, Histories, and Centralized Training

The **history** of player i is $h_i^t = (o_i^0, a_i^0, \dots, o_i^t)$. A **shared policy** with parameters θ maps a history and a player identity to a distribution over actions, $\pi_\theta(a_i^t | h_i^t, i)$. A recurrent implementation compresses the history into a hidden state

$$z_i^t = \text{GRU}_\theta(z_i^{t-1}, o_i^t, a_i^{t-1}), \quad (4)$$

using gated recurrent units [6]. In **centralized training with decentralized execution (CTDE)** [7], a critic $V_\phi(s^t)$ may use the full state during training, while at execution each player acts only from its own history.

2.6 Reward Shaping

A shaping term $F(s, s')$ is guaranteed to leave the optimal policy unchanged if it is potential-based, $F(s, s') = \gamma\Phi(s') - \Phi(s)$ [8]. Our individual-level shaping (passes, tackles, shots) is **not** potential-based, so it can in principle change the optimal policy, which we handle by bounding and annealing it (Section 3.1).

2.7 Performance Measures

For a match $\mathcal{M}$ and policy parameters θ , the **score** (win rate with draws counted as half) is

$$W(\theta; \mathcal{M}) = \Pr[G_T > 0] + \frac{1}{2} \Pr[G_T = 0], \quad (5)$$

where the probability is over the environment, the opponent, and the policy’s randomness. The **expected goal difference** is $\mathbb{E}[G_T]$.

3 Problem Formulation

3.1 Match Model

A **match configuration** is $\mathcal{M} = (f, \mathcal{G}, \sigma, r)$, consisting of our formation f , an opponent formation process $\mathcal{G}$, a skill index σ , and a vision radius r .

State. $s^t = (x_{1:5}^t, y_{1:5}^t, b^t, \beta^t, g^t, t)$.

Actions. Each $a_i^t \in \mathcal{A}$, a finite set containing movement, hold-shape, pass-to-teammate, dribble, shoot, and tackle actions, with invalid actions masked.

Observation. Player i observes

$$o_i^t = (x_{1:5}^t, \beta^t, f, i, \{(y_j^t, v_{ij}^t)\}_{j \in \mathcal{M}}, (b^t, v_{ib}^t)), \quad (6)$$

where $v_{ij}^t = v(y_j^t, x_i^t, r)$ and $v_{ib}^t = v(b^t, x_i^t, r)$, and hidden entries are replaced by zeros. The opponent formation g^t , the opponent’s mode, and σ are *not* part of o_i^t .

Opponent vision. The opponent’s decisions are computed from what each opponent player sees, with $\rho_j = \infty$ if $j = \sigma$ and $\rho_j = r$ otherwise.

Reward. With a goal-based team term and a bounded shaping term,

$$R^t = \underbrace{\mathbb{I}[\text{goal for us}] - \mathbb{I}[\text{goal against}]}_{R_{\text{team}}^t} + \lambda_t \phi(s^t, \mathbf{a}^t) + \mathbb{I}[t = T] \eta \text{sign}(G_T), \quad (7)$$

with shaping term ϕ (passes, tackles, shots on target, turnovers), annealed weight λ_t with $\lambda_t \rightarrow 0$ during training, outcome bonus $\eta > 0$, and a per-episode cap $\sum_t \lambda_t |\phi(s^t, \mathbf{a}^t)| \leq C$.

Objective.

$$J(\theta; \mathcal{M}) = \mathbb{E}_{\pi_\theta, \mathcal{M}} \left[\sum_{t=0}^T \gamma^t R^t \right]. \quad (8)$$

3.2 Q1: Learning to Play

The opponent formation is constant, $g^t \equiv g$, and $\sigma = 0$. Given a distribution $\mathcal{D}_1$ over pairs (f, g) and radii r , find

$$\theta^* \in \arg \max_{\theta} \mathbb{E}_{(f, g, r) \sim \mathcal{D}_1} [J(\theta; (f, g, 0, r))], \quad (9)$$

and report the **formation matrix** $W_{fg} = W(\theta^*; (f, g, 0, r))$ for each pair, together with the baselines $W(\text{random})$ and $W(\text{heuristic})$.

3.3 Q2: Detecting and Adapting to a Formation Change

Switching process. The opponent formation is piecewise constant, $g^t = g_k$ for $t \in [\tau_k, \tau_{k+1})$, with switch times $\tau_1 < \tau_2 < \dots$ and targets $g_k \in \mathcal{F}$. Three modes define $\mathcal{G}$:

- **Scheduled:** one switch at a fixed time τ_1 to a fixed target.
- **Random:** $\tau_{k+1} - \tau_k \geq \delta$ and g_k drawn uniformly from $\mathcal{F}$.
- **Reactive:** $g_k = B(f)$, where $B : \mathcal{F} \rightarrow \mathcal{F}$ is a counter table built in Phase 0.

Training problem. With a training distribution $\mathcal{D}_2^{\text{train}}$ over $(\mathcal{G}, r)$ (with $\sigma = 0$),

$$\theta_{\text{rec}} \in \arg \max_{\theta} \mathbb{E}_{(\mathcal{G}, r) \sim \mathcal{D}_2^{\text{train}}} [J(\theta; (f, \mathcal{G}, 0, r))], \quad (10)$$

evaluated on a test distribution $\mathcal{D}_2^{\text{test}}$ whose switch times and (from, to) formation pairs are disjoint from training.

Compared policy classes: (i) memoryless, $\pi_{\theta}(a \mid o_i^t, i)$; (ii) recurrent, $\pi_{\theta}(a \mid z_i^t, i)$; (iii) **oracle**, $\pi_{\theta^{\text{or}}}(a \mid o_i^t, g^t, i)$, given the true opponent formation.

Regret against the oracle.

$$\text{Reg}(\theta) = W(\theta^{\text{or}}; \mathcal{M}) - W(\theta; \mathcal{M}). \quad (11)$$

Detection. A probe $q_{\psi}(\hat{g}^t \mid z_i^t)$ is trained to predict the opponent formation from the recurrent hidden state. For the k -th switch, the **detection delay** is

$$\Delta_{\text{det}}(k) = \min\{t - \tau_k : t \geq \tau_k, \hat{g}^t = g_k\}. \quad (12)$$

Adaptation lag. Let $\bar{G}_{[a,b]}$ denote the average goal difference per step over steps a to b , and w a window length. With tolerance $\varepsilon \geq 0$,

$$\Delta_{\text{ad}}(k) = \min\{t - \tau_k : t \geq \tau_k, \bar{G}_{[t, t+w]} \geq \bar{G}_{[\tau_k - w, \tau_k]} - \varepsilon\}. \quad (13)$$

Objectives of the study: estimate $\mathbb{E}[\Delta_{\text{det}}]$, $\mathbb{E}[\Delta_{\text{ad}}]$, $\text{Reg}(\theta)$, and the dependence of all three on r .

3.4 Q3: Adapting to an Advantaged Opponent Player

Skill. One opponent player $\sigma \in \{1, \dots, 5\}$ has $\rho_{\sigma} = \infty$, so it can see all players and passing lanes and can attempt interceptions anywhere. The slot type of σ (defender, midfielder, forward) is varied.

Policies.

- θ_{base} : trained with $\sigma = 0$ only.
- θ_{mix} : trained with σ drawn from a distribution $\mathcal{D}_{\sigma}$ over $\{0, 1, \dots, 5\}$.
- θ_{or} : like θ_{mix} but also given σ as an input (oracle).

Performance drop caused by the advantage.

$$D(\theta, \sigma) = W(\theta; (f, g, 0, r)) - W(\theta; (f, g, \sigma, r)). \quad (14)$$

Recovery ratio achieved by adaptation:

$$\kappa(\sigma) = \frac{W(\theta_{\text{mix}}; \sigma) - W(\theta_{\text{base}}; \sigma)}{D(\theta_{\text{base}}, \sigma)}, \quad \kappa = 1 \text{ means full recovery.} \quad (15)$$

Detection. A probe $q_{\psi}(\hat{\sigma} \mid z_i^t)$ predicts the skill index from the recurrent hidden state, with accuracy Acc_{σ} .

Objectives of the study: estimate $D(\theta_{\text{base}}, \sigma)$, $\kappa(\sigma)$, Acc_{σ} , and their dependence on the slot of σ .

4 Proposed Solutions

4.1 Environment and Heuristic Opponent (Phase 0)

- Implement the 2D environment of Section 3.1: continuous positions, ground-ball passing with interception geometry, dribbling and tackling, distance-based shooting, positional control rating, and limited vision.

- Implement the heuristic opponent: pass if a free teammate is ahead, shoot if unmarked in the shooting zone, otherwise dribble; zonal defending; vision-limited reasoning; the skilled variant that reasons over the full pitch; and the three formation-switching modes.
- **Calibrate** all numeric constants (pitch, speeds, radius values, control-rating constants, shooting and tackle parameters) by running heuristic-vs-heuristic matches, checking that they are balanced and that a random policy loses clearly.
- **Result to report:** a heuristic-vs-heuristic formation win matrix, which also defines the counter table B used by the reactive opponent.

4.2 Learning Algorithm

- Parameter-shared PPO [9] with generalized advantage estimation [10], a centralized critic and decentralized actors [11, 7]. Two actor variants: feed-forward and GRU [6].
- Reward with the two-tier structure of Section 3.1, using an annealed and capped shaping term. Run a short diagnostic training first to check that shaping return does not grow while win rate stays flat, and reweight if it does.
- Curriculum: one formation on both sides first, then all formations, then varying r , with widening points marked on learning curves.
- Multiple training seeds per configuration (number TBD), with fixed evaluation seeds shared across policies for paired comparison.

4.3 Plan and Expected Results for Q1

- **Experiments:** train against the heuristic for each formation pair, at several radii.
- **Results by final submission:**
 - Learning curves (mean and standard deviation across seeds) against the random and heuristic baselines.
 - The formation matrix W_{fg} of the learned team, compared with the heuristic-only matrix from Phase 0.
 - Behaviour statistics: possession share, pass count and completion, shots, and team spread, showing what the team learned.
- **Hypothesis:** the learned team beats random clearly and matches or beats the heuristic, with a performance drop as r shrinks.

4.4 Plan and Expected Results for Q2

- **Experiments:** train and test the feed-forward, recurrent, and oracle policies against the three switching modes, at several values of r , with held-out switch times and formation pairs.
- **Results by final submission:**
 - Table of W and Reg for each (mode, policy) pair.
 - Detection delay Δ_{det} from the probe, and adaptation lag Δ_{ad} , per mode and per radius.
 - Curves of goal-difference rate around a switch (aligned at τ_k) for the three policies.
- **Hypothesis:** the recurrent policy outperforms the memoryless one, and its gap to the oracle grows as r shrinks. Reactive switching is the hardest mode.

4.5 Plan and Expected Results for Q3

- **Experiments:** evaluate θ_{base} , θ_{mix} , and θ_{or} against skilled opponents, with the skill placed on a defender, a midfielder, and a forward in turn.
- **Results by final submission:**
 - Performance drop $D(\theta_{\text{base}}, \sigma)$ per slot type.
 - Recovery ratio $\kappa(\sigma)$ of the mixed-training policy, and the gap to the oracle.
 - Detection accuracy Acc_σ of the probe, and a behavioural comparison (for example, whether passes avoid lanes visible to the skilled player).
- **Hypothesis:** the advantage costs the unaware team measurable performance, mixed training recovers part of it, and the skill matters most in the slot where the full-vision information is most useful (to be determined empirically).

4.6 Evaluation Protocol

- Report the mean and standard deviation over training seeds, and many evaluation matches per condition (numbers TBD), with paired evaluation seeds across policies.
- Always run the random-action baseline before trusting any trained win rate.
- Track shaping return against win rate throughout training as the reward-hacking indicator.
- A negative or partial result is a valid finding, provided it is quantified (for example, “the team cannot detect reactive switches at small r ”).

5 Contribution

The work is split into three areas, one lead per area, with every member reviewing and contributing to the shared tasks. The assignment below is a proposed split and can be adjusted as the project progresses.

5.1 Task Assignment

Table 2: Proposed task assignment.

Member	Primary responsibility	Tasks
Arpan Gupta	Environment and opponent (Section 4.1)	Implement the 2D football environment (passing and interception, dribbling and tackling, shooting, positional control rating, limited vision). Implement the heuristic opponent, including the three formation-switching modes and the full-vision skilled player. Run the Phase 0 calibration (heuristic-vs-heuristic matches, formation win matrix, counter table).
Deep Buha	Learning and Q1 (Sections 4.2 and 4.3)	Implement the parameter-shared PPO team with a centralized critic, in feed-forward and GRU versions. Design and tune the reward function, including annealed and capped shaping and the reward-hacking monitor. Run the curriculum training and the Q1 experiments (learning curves, formation matrix, baselines, behaviour statistics).
Ramji Purwar	Adaptation experiments, Q2 and Q3 (Sections 4.4, 4.5 and 4.6)	Implement the oracle variants and the probes for opponent formation and skill detection. Run the Q2 experiments (switching modes, held-out schedules, detection delay, adaptation lag, regret) and the Q3 experiments (skill slot, performance drop, recovery ratio). Build the evaluation pipeline and produce the tables and figures.

References

- [1] Hiroaki Kitano, Minoru Asada, Yasuo Kuniyoshi, Itsuki Noda, and Eiichi Osawa. RoboCup: The robot world cup initiative. In *Proceedings of the First International Conference on Autonomous Agents*, 1997.
- [2] Karol Kurach et al. Google research football: A novel reinforcement learning environment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020. arXiv:1907.11180.
- [3] Yan Song, He Jiang, Haifeng Zhang, Zheng Tian, Weinan Zhang, and Jun Wang. Boosting studies of multi-agent reinforcement learning on google research football environment: the past, present, and future. *arXiv preprint arXiv:2309.12951*, 2023.
- [4] Daniel S. Bernstein, Robert Givan, Neil Immerman, and Shlomo Zilberstein. The complexity of decentralized control of Markov decision processes. *Mathematics of Operations Research*, 27(4), 2002.
- [5] Frans A. Oliehoek and Christopher Amato. *A Concise Introduction to Decentralized POMDPs*. Springer, 2016.

- [6] Kyunghyun Cho et al. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.
- [7] Ryan Lowe et al. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [8] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the International Conference on Machine Learning (ICML)*, 1999.
- [9] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [10] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.
- [11] Chao Yu et al. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.